HACK-X 2026 · 35-HOUR HACKATHON · MJCET

KisanSaathi

(किसान साथी)

Empowering Farmers with
Direct Mandi Access, Computer Vision Quality Grading & Cluster Logistics



THEME

Agriculture, FoodTech & Rural Development

PROBLEM SOLVED

Eliminates intermediaries and subjective mandi quality exploitation

Reduces prohibitive freight costs for smallholder farmers

TEAM FOREVERKNIGHTS

Gulam Waheed Mujtaba – Lead Architect & Full Stack Lead
Shaik Mohammed Ateeq – Cloud Infrastructure & AI Systems Engineer
Malik Samiullah Khan – UI/UX & Agricultural Tech Specialist

LIVE PROTOTYPE

Full-featured PWA with bilingual AI assistant, Leaflet GPS tracking & escrow wallet

KISANSAATHI SOLUTION OVERVIEW

5-STEP CORE WORKFLOW

01

Instant Crop Listing

Farmer uploads produce details or photo scan

02

Automated AI Grading

Computer Vision predicts grade (Elite/Premium/Classic) & price range

03

Cluster Aggregation

Pooling sub-500kg harvests; 38% logistics cost reduction (Warangal Agri-Cluster #12)

04

Direct Buyer Negotiation

Transparent counter-offer desk connecting farmers & bulk buyers (Hyderabad & Secunderabad)

05

Guaranteed Escrow Settlement

DBT payout within 24h, real-time GPS transit tracking

TRADITIONAL MANDI VS KISANSAATHI

FEATURE	TRADITIONAL MANDI	KISANSAATHI
Middlemen Commission	12% - 18% loss	0% (Direct Escrow)
Quality Assessment	Subjective visual bias	Computer Vision automated
Freight Economics	Individual prohibitive cost	Shared cluster pooling (38% savings)
Payment Settlement	15-45 days delayed credit	24-hour DBT bank transfer

TECHNICAL ARCHITECTURE & STACK

01
Farmer Input
Image / Crop Data

→

02
AI Quality Engine
OpenCV / TensorFlow.js

→

03
Smart Contract Ledger
Immutable Loan Registry

→

04
Real-Time Marketplace
Negotiation Desk

→

05
Logistics & GPS Transit
Leaflet Maps

TECH STACK

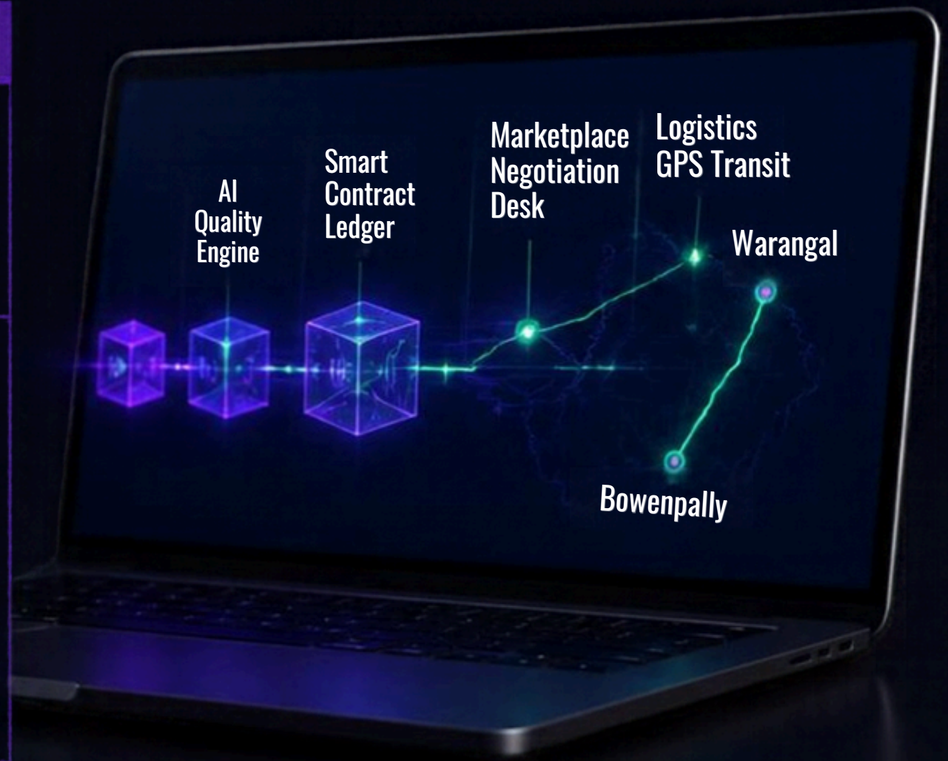
Frontend	Progressive Web App (HTML5, Tailwind CSS, Lucide Icons)
Backend & Engine	Node.js REST API, zero-dependency static streaming server
AI & Computer Vision	Client-side grain defect detection, grade classification, price prediction
Bilingual Assistant	English/Hindi NLP with Web Speech Synthesis API
Mapping & Geo-Tracking	Leaflet.js with live Telangana waypoint navigation (Warangal to Bowenpally)

KEY INNOVATIONS

OFFLINE-CAPABLE MOBILE APP SIMULATION

BLOCKCHAIN-BACKED AGRICULTURAL CREDIT SCORE & 4% SUBSIDIZED LOAN LEDGER

785/900
CREDIT RATING



FEASIBILITY, RISKS & DEVELOPMENT ROADMAP

VIABILITY PILLARS

TECHNICAL
Lightweight web architecture
Client-side AI
Minimal bandwidth for rural 4G

ECONOMIC
Zero listing fee
Monetized via 1.5% bulk corporate transaction fee

OPERATIONAL
Integrated with Telangana Warehousing Corp godowns & authorized Soil Testing Labs



RISK & CHALLENGE MITIGATION

CHALLENGE		MITIGATION
Rural digital literacy & language barriers	→	Bilingual voice assistant & single-tap OTP login
Fragmented harvest volumes <500kg	→	Automated Cluster Aggregation Engine pooling farmers
Trust in electronic payments	→	Verified escrow releases & blockchain transaction ledger

DEVELOPMENT STATUS & ROADMAP

CURRENT
Working bilingual prototype
AI crop grader
Escrow wallet
Telangana lab directory

→

NEXT PHASE
Integration with Government e-NAM APIs & satellite vegetation index (NDVI)

IMPACT & BENEFITS

Value for Every Stakeholder

SMALLHOLDER FARMERS

20%-28%

increase in net income

Eliminates arbitrary quality deductions

BULK BUYERS & MILLERS

Traceable farm-gate sourcing

Verified moisture & quality certificates

Automated invoicing

AGRICULTURE DEPARTMENT

Real-time district-level yield data

Crop disease tracking

DBT subsidy disbursment



ECONOMIC

₹4,200 – ₹6,500

additional earnings per metric ton of Telangana Sona (RNR 15048)

OPERATIONAL

₹3.80/KG → ₹2.35/KG

freight cost cut via consolidation

SOCIAL & ETHICAL

inclusive digital access for Warangal, Medchal, Secunderabad & Nizamabad farming communities

RESEARCH & REFERENCES

EVIDENCE & MARKET VALIDATION

NITI Aayog: Agricultural Marketing & Farmer Friendly Reforms Index (intermediary margin losses)

Directorate of Economics & Statistics Telangana: Crop productivity reports (Telangana Sona & Cotton)

Ministry of Agriculture & Farmers Welfare: Soil Health Card Scheme & PMFBY guidelines

TECHNOLOGY & DOMAIN FRAMEWORKS

ICAR-GRIDA & PITSAU Rajendranagar: Soil nutrient thresholds & agronomic guidelines

IEEE / ScienceDirect: Computer Vision in grain quality inspection & defect estimation

OUR CORE POSITIONING

Traditional portals merely list contact numbers. KisanSaathi **standardizes quality** with AI, **eliminates freight barriers** through clustering, and **guarantees payment** with escrow.

